"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 24UECC0053

B.E. III (ECE)

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DCS

THIRD B.E. EXAMINATION - 2026
ELECTRONICS & COMMUNICATION ENGINEERING
(VI SEMESTER CBCS)
(For MAIN/ATKT/Ex Students)
6ECE41A(ECE)/6ECC41A(ECE): DIGITAL COMMUNICATION SYSTEMS (ECE/ECC)

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Derive the relationship between joint probability density function f(x,y) and the marginal density functions fX(x) and fY(y). If f(x,y) = 2 for $0 \leq x \leq y \leq 1$, find the marginal densities and determine if X and Y are independent. 7

(b) What is Multiplexing? Explain the difference between Frequency Division Multiplexing and Time Division Multiplexing. In a TDM system, 5 voice signals, bandwidth 4 kHz each are sampled at Nyquist rate and quantized using 8 bits. Find the final bit rate. 7

2. (a) A PCM system uses 8-bit quantization and a sampling rate of 10 kHz. Calculate (a) the bit rate, (b) the bandwidth required for transmission, and (c) the maximum SQNR in dB assuming a full-scale sinusoidal input. 7

(b) Describe Adaptive Delta Modulation. How does it overcome the limitations of linear delta modulation? Draw the block diagram of an ADM system and explain its operation. 7

3. (a) Explain Gaussian noise and derive its statistical properties. 7

(b) Explain the generation and detection of Quadrature Phase Shift Keying. Draw its constellation diagram and state its advantages over BPSK. 7

4. (a) Derive the condition for slope overload in a delta modulator. If a delta modulator is designed to track a signal $m(t) = A \sin(2\pi fmt)$, find the minimum step size required to avoid slope overload. 7

(b) Analyze performance of delta modulation in noisy environments. 7

5. (a) Explain the concept of white noise. Is white noise physically realizable? Why or why not? A white noise process with PSD N0/2 is passed through an ideal bandpass filter of bandwidth B centered at fc. Find the autocorrelation function of the output noise. 7

(b) Describe the complete digital communication system with detailed functional analysis. 7

6. (a) Derive the matched filter response and prove that is maximizes SNR. 7

(b) Analyze the output noise of a system when white noise is passed through an LTI system. 7

7. (a) Derive the probability of error for BPSK in AWGN channel. 8

(b) Compare coherent and non-coherent FSK systems in terms of error performance. 6